

Sparse Zero-Diagonal Compact Operators Are Compact Commutators

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Status and source question

This is a `candidate_partial_likely_valid` result for Sasmita Patnaik and Rahul Sethi, *On Simultaneous Triangularization of Matrices and Quasinilpotency of Commutator of Compact Operators*, arXiv:2401.16916 [1]. The unnumbered question on PDF page 2 asks:

Is every compact quasinilpotent operator a commutator of compact operators?

from a more than half-a-century old open question of Percy and Topping posed in 1971: Is every compact operator a commutator of compact operators? In 2017, Ken Dykema and Amudhan Krishnaswamy Usha proved that every compact nilpotent operator is a commutator of compact operators [3] Theorem 3.2]. Their work involved the following structure theorem for nilpotents which they modified for entries to act of the same space in [3] Lemma 2.2]: for every nilpotent $T \in B(\mathcal{H})$, i.e., $T^n = 0$ for some $n > 1$, there exists a decomposition $\mathcal{H} = \oplus_{i=1}^n \mathcal{H}_i$ of the Hilbert space $\mathcal{H}$ into the direct sum of an orthogonal family of subspaces $\mathcal{H}_1, \mathcal{H}_2, \dots, \mathcal{H}_n$ such that the matrix of T corresponding to this decomposition has the upper triangular form [6] Theorem 1]. This finite block matrix model allowed them to deal with only finitely many operator equations to construct appropriate compact operators C and Z for which $CZ - ZC = T$, though the computations are quite intricate.

In the case of operators acting on finite-dimensional Hilbert space, where spectrum of an operator being $\{0\}$ implies that the operator must be nilpotent. In contrast, there are operators acting on infinite-dimensional Hilbert spaces whose spectrum is $\{0\}$, but they are not nilpotent. Such operators are called quasinilpotent operators.

For the class of compact quasinilpotent operators that are not nilpotent, we ask the same question as posed earlier for compact nilpotent operators. Our study is centered around the following question.

Question. *Is every compact quasinilpotent operator a commutator of compact operators?*

Unlike the case of nilpotent operator, one does not have a finite matrix structure for compact quasinilpotent operators, so addressing such a question in full generality seems intractable! In order to make any headway, alternatively, we pose the following question:

Question 1. *Determine sufficient condition(s) on compact operators C and Z for which the commutator $CZ - ZC$ is a compact quasinilpotent operator. In other words, when is the commutator of compact operators a quasinilpotent operator?*

In the literature, some sufficient conditions on compact operators C and Z are provided to answer Question 1. For instance, if the pair of compact operators $\{C, Z\}$ is simultaneously triangularizable, then $[C, Z]$ is quasinilpotent [10] Theorem 7.3.3]; if every word in CZ and ZC is quasinilpotent, then $[C, Z]$ is quasinilpotent (a consequence of Turovskii's theorem [10] Corollary 8.1.14] applied to CZ and

The theorem below answers this for a concrete sparse subfamily, and in fact does not need quasinilpotence. The general question remains open.

Fix an orthonormal basis $(e_i)_{i \in \mathbb{N}}$ of a separable Hilbert space. If $T = (t_{ij})$, where $t_{ij} = \langle Te_j, e_i \rangle$, its *undirected support graph* has vertex set $\mathbb{N}$ and an edge $\{i, j\}$, $i \neq j$, exactly when $t_{ij} \neq 0$ or $t_{ji} \neq 0$.

Proof intuition

A commutator with a diagonal operator is entrywise division by differences of diagonal entries. Small differences are the usual obstruction. Bounded support degree gives a finite proper coloring. Put the colors on finitely many distinct complex rays and give vertex i a radial length equal to the square root of the largest matrix entry touching it. Compactness makes these lengths tend to zero, so the diagonal factor is compact. Distinct rays keep the denominator on every support edge comparable to the larger radial length; after division, the new entry is bounded by the smaller radial length. The bounded degree then turns entrywise tail decay into operator-norm tail decay by the Schur test, proving that the second factor is compact.

This mechanism fails for the currently known universal block-tridiagonal forms because their scalar support degrees grow with the block sizes.

Main theorem

Theorem 1 (bounded-degree support). *Let $T \in \mathcal{K}(\mathcal{H})$ have zero diagonal in an orthonormal basis (e_i) . Suppose its undirected support graph has degree at most $\Delta < \infty$. Then there are compact operators $A, B \in \mathcal{K}(\mathcal{H})$ such that*

$$T = [A, B] = AB - BA.$$

Moreover, A may be chosen diagonal and B may be chosen with matrix support contained in that of T .

Proof. Write G for the support graph. A greedy coloring of a countable graph of maximum degree Δ gives a proper coloring $c : \mathbb{N} \rightarrow \{0, \dots, \Delta\}$. Choose distinct unit complex numbers $\zeta_0, \dots, \zeta_\Delta$. Since there are only finitely many distinct rays, the number

$$\delta = \min_{p \neq q} \min_{0 \leq s \leq 1} |\zeta_p - s\zeta_q|$$

is strictly positive. Consequently, for $r, u \geq 0$ and $p \neq q$,

$$|r\zeta_p - u\zeta_q| \geq \delta \max\{r, u\}. \quad (1)$$

For each vertex put

$$\rho_i = \sup_{j \neq i} \max\{|t_{ij}|, |t_{ji}|\}, \quad a_i = \sqrt{\rho_i}, \quad \lambda_i = a_i \zeta_{c(i)}.$$

(The supremum is zero at an isolated vertex.) Compactness of T and T^* gives

$$\rho_i \leq \max\{\|Te_i\|, \|T^*e_i\|\} \longrightarrow 0,$$

because (e_i) is weakly null. Hence the diagonal operator $Ae_i = \lambda_i e_i$ is compact.

For $i \neq j$, define

$$b_{ij} = \begin{cases} \frac{t_{ij}}{\lambda_i - \lambda_j}, & t_{ij} \neq 0, \\ 0, & t_{ij} = 0, \end{cases} \quad b_{ii} = 0.$$

Whenever $t_{ij} \neq 0$, the vertices i, j are adjacent and hence differently colored. Both ρ_i and ρ_j dominate $|t_{ij}|$. By (1),

$$|b_{ij}| \leq \delta^{-1} \frac{|t_{ij}|}{\max\{a_i, a_j\}} \leq \delta^{-1} \min\{a_i, a_j\}. \quad (2)$$

There are at most Δ nonzero entries in each row and column of $B = (b_{ij})$. Thus the Schur test first shows that this matrix defines a bounded operator.

It also shows compactness. Given $\varepsilon > 0$, choose M so that $a_i < \varepsilon\delta/\Delta$ for $i > M$ (with the evident harmless change when $\Delta = 0$). The first M vertices have only finitely many neighbors; choose $N \geq M$ larger than all their indices. Every support edge with at least one endpoint greater than N then has both endpoints greater than M . By (2), all corresponding entries have modulus less than ε/Δ . The row and column sum bounds in the Schur test give

$$\|B - P_N B P_N\| < \varepsilon,$$

where P_N projects onto $\text{span}\{e_1, \dots, e_N\}$. Hence B is a norm limit of finite-rank operators and is compact.

Finally, the diagonal entries of $[A, B]$ vanish, as do those of T , and for $i \neq j$,

$$[A, B]_{ij} = (\lambda_i - \lambda_j)b_{ij} = t_{ij}.$$

Therefore $T = [A, B]$. □

Consequences and an upgrade

Corollary 2. *Every compact zero-diagonal finite-band matrix is a commutator of two compact operators. In particular, every compact unilateral or bilateral weighted shift is such a commutator.*

Proof. A matrix of bandwidth m has support degree at most $2m$. A weighted shift has support degree at most two. Apply Theorem 1. □

Every compact unilateral weighted shift is quasinilpotent: if its weights tend to zero, its spectral radius is bounded by the supremum of every tail of the weight sequence. Taking all weights nonzero gives a nonnilpotent compact quasinilpotent covered by Corollary 2. Thus the corollary is a literal nonnilpotent subcase of the source question, not merely a reproof of the known nilpotent case.

The same argument has a useful block form.

Corollary 3 (uniform finite blocks). *Let $\mathcal{H} = \bigoplus_{n \geq 1} \mathcal{H}_n$, with $1 \leq \dim \mathcal{H}_n \leq r < \infty$, and let $T = (T_{mn})$ be compact. Suppose the graph joining block vertices m, n when $T_{mn} \neq 0$ or $T_{nm} \neq 0$ has uniformly bounded degree, and*

$$\text{tr}(T_{nn}) = 0 \quad (n \geq 1).$$

Then T is a commutator of two compact operators.

Proof. Every trace-zero matrix has an orthonormal basis with zero diagonal. Indeed, zero is in the convex hull of its eigenvalues and hence in its numerical range, so choose a unit vector x with $\langle Sx, x \rangle = 0$; the compression to $x^\perp$ again has trace zero, and induction completes the basis. Apply this inside each $\mathcal{H}_n$ and take the direct sum of the resulting unitaries. The transformed T has zero scalar diagonal. A scalar vertex is adjacent to at most $r - 1$ vertices in its own block and to at most r times the block-degree outside it, so the scalar support degree is uniformly bounded. Theorem 1 applies, and unitary conjugation transfers the commutator representation back to T . □

Verification, limitations, and novelty status

The included script tests the exact factorization, proper coloring, ray separation, and estimate (2) on 2,900 random finite sparse matrices. All cases passed with operator-norm residual below $2 \cdot 10^{-15}$. This is a regression check only; compactness is proved by the Schur-tail argument above.

Eight focused routes were examined. Finite-rank nilpotent approximation only produces a limit of commutators; a commutator range need not be closed. The diagonal-derivation route succeeded for bounded-degree zero-diagonal support and was upgraded to Corollary 3. It does not extend directly to unbounded degrees. Universal block-tridiagonalization has exponentially growing block sizes, while changing to a zero-diagonal basis does not preserve sparsity. Hardy-type factorization and stabilization of a sum of commutators also fail to give the original operator rather than an amplification. Finally, the usual eigenvalue/trace constraints yield no counterexample for quasinilpotents.

The run indexes and bounded web/arXiv searches used the exact source question and the phrases *zero diagonal commutator of compact operators*, *bounded degree sparse matrix compact commutator*, and *compact weighted shift commutator compact*. They located the general Percy–Topping literature, including the block-tridiagonal analysis of Loreaux–Patnaik–Petrovic–Weiss [2], but no exact theorem with the bounded support-degree hypothesis and two compact factors. Novelty confidence is moderate pending expert review. The source’s question for arbitrary compact quasinilpotents remains open.

Human-review recommendation. Check especially the passage from (2) to norm compactness of B and search older sparse-matrix commutator literature for an equivalent formulation. The proof itself is elementary and otherwise self-contained.

References

- [1] S. Patnaik and R. Sethi, *On Simultaneous Triangularization of Matrices and Quasinilpotency of Commutator of Compact Operators*, arXiv:2401.16916 (2024).
- [2] J. Loreaux, S. Patnaik, S. Petrovic, and G. Weiss, *On commutators of compact operators via block tridiagonalization: generalizations and limitations of Anderson’s approach*, Integral Equations Operator Theory 95 (2023), Article 9; arXiv:2203.11995.